

“Math is Cool” Championships -- 2025-26

5th Grade

Mental Math Solutions

	Answer	Solution
1	76	What is the sum of twenty-five and fifty-one? $25 + 51 = 76$
2	24 [inches]	In inches, what is the perimeter of a square with a side length of six inches? $6 + 6 + 6 + 6 = 24$
3	34	What is fifty percent of sixty-eight? $68 \text{ divided by } 2 = 34$
4	55 [= median]	Charlie scores a fifty, forty-eight, fifty-five, sixty and sixty-two on his most recent math tests. Which score is the median? The median is the number in the middle of a data set.
5	20 [= x]	X plus fourteen equals thirty-four. What is x? $x = 34 - 14 = 20$
6	1 [feet, or foot]	A right triangle has a base of four feet and a height of five feet. A rectangle with a base of ten feet has the same area as the triangle. What is the height of the rectangle in feet? $(4 \times 5)/2 = 10$ $10 \times 1 = 10$
7	42 [= total value]	The letters of the alphabet A through Z are each assigned an integer from one to twenty-six in order. What is the total value of the letters in the word MATH, spelled M-A-T-H, when added together? $M = 13$ $A = 1$ $T = 20$ $H = 8$ $13 + 1 + 20 + 8 = 42$

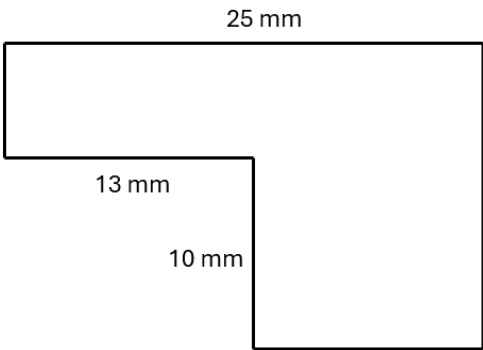
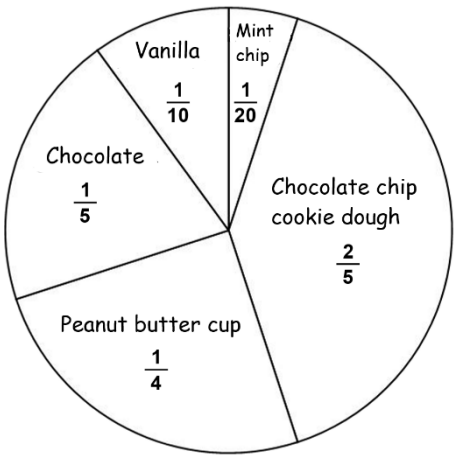
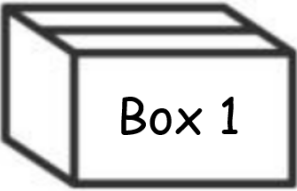
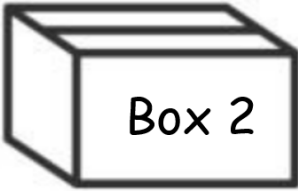
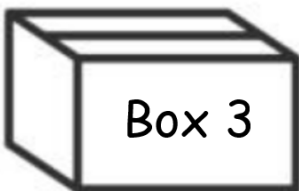
8	224 [= 8th term]	The first two terms of a sequence are two and five. Each of the following terms in the sequence is equal to the sum of all the terms that come before it in the sequence. What is the eighth term in the sequence? 2, 5, 7, 14, 28, 56, 112, 224 Starting with the 4th term, each term is just two times the previous term.
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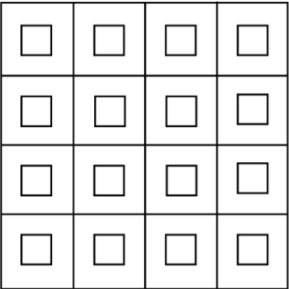
“Math is Cool” Championships -- 2025-26

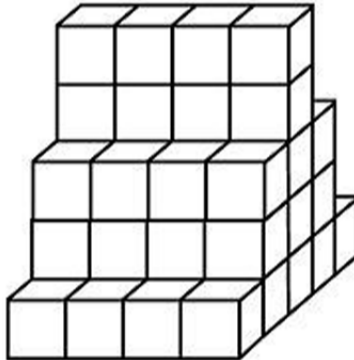
5th Grade

Individual Test Solutions

	Answer	Solution
1	3616	What is 452 multiplied by 8? $452 \times 8 = 3616$
2	31 [=x]	If $x - 15 = 16$, what is the value of x? $x = 15 + 16 = 31$
3	50	What is the place value of the digit 5 in the number 358? The five is in the 10's place.
4	3	As an integer, what is the value of the fraction? $\frac{27}{9}$ Divide 27 by 9 = 3
5	80 [degrees]	A triangle has angle measures of 40° and 60° . In degrees, what is the measure of the third angle? All interior angles of triangles add to 180° .
6	4 [apples]	Delisha has 8 apples and 2 oranges. How many apples are there for every orange? $8/2 = 4$
7	11	What is 10.58 rounded to the nearest whole number? The 5 tells you to round up.
8	12 [cm]	A square has a side length of 3 centimeters. In centimeters, what is the perimeter of the square? A square has 4 sides. $3+3+3+3 = 12$.
9	46 [= mode]	What is the mode of the data set: {40, 45, 46, 46, 50, 52}? # that occurs most.
10	18 [crayons]	Ginger has 3 boxes of crayons. Each box has 6 crayons inside. How many crayons does she have in total? $3 \times 6 = 18$
11	7 [marbles]	Sophie has some marbles. Her friend gives her 5 more, and now she has 12. How many marbles did Sophie have to start with? $x + 5 = 12, x = 7$
12	8 [= a factor]	Which of the following is a factor of 32? 3, 6, 8, 10, 12, 17 The only number in the list that divided evenly into 32 is 8.
13	54 [inches]	How many inches are there in 4.5 feet? $4.5 \times 12 = 54$

14	270 [square mm, mm ²]	In square millimeters, what is the area of the figure shown? If the corner were not missing, it would be $25 \times 16 = 400$. Subtract off the missing corner, which is $13 \times 10 = 130$. $400 - 130 = 270$. 
15	275	Find the missing number in the sequence: 11, 55, _____, 1375, 6875, ... Each number in the sequence is being multiplied by 5.
16	60 [%]	The following graph shows the favorite ice cream flavors of Mrs. Schuh's 5th grade students. What percentage of students have either chocolate or chocolate chip cookie dough as their favorite flavor? Chocolate + chocolate chip cookie dough = $\frac{1}{5} + \frac{2}{5} = \frac{3}{5} = \frac{6}{10} = 60\%$. 
17	162 [minutes]	If a movie starts at 2:05 p.m. and ends at 4:47 p.m. the same afternoon, how many minutes did the movie last? $2:05 - 4:05 = 2 \text{ hours} = 120 \text{ minutes}$. $4:05 - 4:47 = 47 - 5 = 42 \text{ minutes}$. $120 + 42 = 162$.
18	3 [= box number]	Each of the boxes has one true statement and one false statement written below them. Which box number contains the prize?  Box 1 1. The prize is not in here. 2. The prize is not in Box 3.  Box 2 1. The prize is in Box 2. 2. The prize is not in Box 1.  Box 3 1. The prize is not in here. 2. The prize is not in Box 1. Assume it is in Box 1, therefore statement 1 is false and statement 2 is true. However, this makes both statement for Box 2 false. Assume it is in Box 2, therefore statement 1 is true but statement 2 is also true. Assume it is in Box 3, therefore statement 1 is false and statement 2 is true. For Boxes 1 and 2, each of them now have one true and one false statement.

19	52 [dollars]	You have 3 ten-dollar bills, 4 five-dollar bills, and 2 one-dollar bills. How many dollars do you have in total? $3(10)=30$ $4(5)=20$ $2(1)=2$ $30+20+2=52$
20	12 [= GCF]	What is the greatest common factor of 48 and 60? $12 \times 4 = 48$ $12 \times 5 = 60$
21	50 [snack packs]	A 5th grade class is organizing a fundraiser. They decide to sell snack packs to raise money for a field trip. Each snack pack contains the following: 2 juice boxes, 1 granola bar, and 3 fruit snacks. If the class has 120 juice boxes, 60 granola bars, and 150 fruit snacks, what is the maximum number of complete snack packs the students can make with the supplies they have? Juice boxes: $120 \div 2 = 60$ Granola bars: $60 \div 1 = 60$ Fruit snacks: $150 \div 3 = 50$
22	9 [= positive difference]	What is the positive difference between the largest and the smallest prime factors of 220? $220 = 2^2 \cdot 5 \cdot 11$ $11 - 2 = 9$
23	46 [squares]	How many squares of any size are in the following figure? Smallest - 16 Next smallest - 16 $2 \times 2 = 9$ $3 \times 3 = 4$ $4 \times 4 = 1$ Total = 46 
24	10 [books]	Hyrum has 72 books that he wants to organize evenly on 6 shelves. After he is done organizing the books, he has 12 books left over that he will donate. How many books did he put on each of the 6 shelves? $72 - 12 = 60$ 60 books divided by 6 shelves equals 10 books on each shelf.
25	10 [goals]	Brooklyn recorded the number of goals in each of her 5 soccer games: 2, 3, 4, 5, 6. How many goals does she need to score in her 6th game to have an average of 5 goals per game overall? $2+3+4+5+6 = 20$ To have an average of 5 goals in 6 games, she needs a total of 30 goals, therefore she needs to score 10 goals.

26	424 [sq inches]	The following figure is made up of 52 cubes of edge length 2 inches, which have been glued together into a symmetric shape. In square inches, what is the total surface area of the figure? Each square = $2 \times 2 = 4$ square inches. Bottom = $4 \times 5 = 20$ squares. Right = 13 squares. Left side = 13 squares. Each top facing side = 4 squares $\times$ 5 tops = 20 squares. Each side facing (front or back) side = 4 squares $\times$ 10 sides = 40 squares. Total squares = 106. $106 \times 4 = 424$ sq. inches.																																																				
27	15 [cups]	A cookie recipe requires $3 \frac{1}{2}$ cups of flour for every $1 \frac{1}{2}$ cups of sugar. Harlen wants to triple the recipe to make enough for a class party. How many total cups of sugar and flour combined will be needed? $(3 \times 3 \frac{1}{2}) + (3 \times 1 \frac{1}{2}) = 9 + 3\frac{1}{2} + 3 + 3\frac{1}{2} = 12 + 6\frac{1}{2} = 12 + 3 = 15$ total cups of flour and sugar																																																				
28	59 [= 20 th number]	What is the 20th number in this pattern? 2, 5, 8, 11, ... 2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32, 35, 38, 41, 44, 47, 50, 53, 56, 59.																																																				
29	32[%]	There are 16 integers that are a multiple of 5 or 7 (or both). $16/50 = 32/100 = 32\%$. <table border="1" data-bbox="443 1110 1261 1457"><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td></tr><tr><td>21</td><td>22</td><td>23</td><td>24</td><td>25</td><td>26</td><td>27</td><td>28</td><td>29</td><td>30</td></tr><tr><td>31</td><td>32</td><td>33</td><td>34</td><td>35</td><td>36</td><td>37</td><td>38</td><td>39</td><td>40</td></tr><tr><td>41</td><td>42</td><td>43</td><td>44</td><td>45</td><td>46</td><td>47</td><td>48</td><td>49</td><td>50</td></tr></table>			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
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41	42	43	44	45	46	47	48	49	50																																													
30	16 [cats]	An animal shelter has cats and dogs. There are four more cats than dogs in the shelter. If there are a total of 28 animals, how many of them are cats? $x + x + 4 = 28$ $2x + 4 = 28$ $2x = 24$ $x = 12$ dogs $28 - 12 = 16$, so there are 16 cats.																																																				

31	45	In the following pattern of dates, what would complete the last date? Enter your answer as a 2-digit integer, where the two digits would fill in the two blanks. January 2, 2021 February 8, 2082 March 9, 2093 April 1, 2014 May 4, 20__ They are all palindromes: 1/2/21 2/8/82 3/9/93 4/1/14 5/4/45
32	25 [= A + B]	Forty cards have been labeled with the integers 1 through 40. Four different cards are randomly selected, and placed in order of their selection from left to right. The probability that the number on each card in order from left to right is greater than the previous number can be written as a simplified fraction A/B. What is A + B? Any 4 cards can be ordered in $4! = 24$ ways. There is only 1 way that the cards are in ascending order. $P = 1/24$, therefore $A + B = 25$.
33	96 [whole pieces]	An iron bar is 2.9 meters long. If it is cut into pieces that are 3 centimeters long, how many whole pieces can be made? 2.9 meters = 290 centimeters. $290/3 = 96 \frac{2}{3}$, therefore 96 whole pieces can be made.
34	333 [= mean]	The digits 2 and 4 can be used to create 8 distinct three-digit positive integers, using each digit (2 or 4) from 0 to 3 times. What is the mean of these 8 integers? The integers are 222, 422, 242, 224, 244, 424, 442, 444. The sum of the 8 integers is 2664. $2664/8 = 333$.
35	80 [= 33 rd term]	In a geometric sequence of positive numbers, if the 21st term is 500 and the 39th term is 32, what is the 33rd term? Let r be the common ratio of the sequence, then $500 \cdot r^{18} = 32 \rightarrow r^{18} = 32/500 = 8/125 \rightarrow (r^{18})^{(1/3)} = (8/125)^{(1/3)} \rightarrow r^6 = 2/5 \rightarrow (r^6)^2 = (2/5)^2 \rightarrow r^{12} = 4/25$. The 33rd term will be $500 \cdot r^{12} = 500 \cdot 4/25 = 80$
36	252 [3-digit integers]	How many positive three-digit integers contain the digit 9 at least once? Case study: Three 9's: 1 way (999) Two 9's: 26 ways (9 ways for 99_, 9 ways for 9_9, 8 ways for _99) One 9: 225 ways (81 ways for 9__, 72 ways for _9_, 72 ways for __9). $1 + 26 + 225 = 252$

37

324 [= A]

Two concentric circles contain a horizontal line which is 36 cm long, and that touches the outside edges of the outer circle, and is tangent to the inner circle. The shaded area of the figure is $A\pi \text{ cm}^2$.

What is the value of A?

Let R = radius of big circle, and r = radius of inner circle.

$$\text{Area big} = \pi R^2$$

$$\text{Area inner} = \pi r^2$$

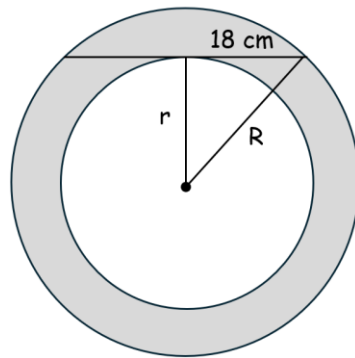
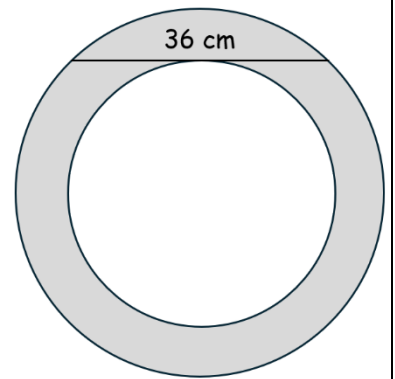
$$\text{Shaded area} = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$$

$$\text{Also, } r^2 + 18^2 = R^2$$

$$R^2 - r^2 = 18^2 = 324$$

$$\text{Therefore, Shaded area} = 324 \pi.$$

$$A = 324$$



38	11 [= sum]	In the following puzzle, place the digits 1 through 9 in the outlined empty squares, using each digit exactly once, in order to make the three horizontal and three vertical equations true. Calculations are done from left to right and top to bottom, NOT using the usual "order of operations". For example: $3 + 4 \times 2 = 7 \times 2 = 14$. What is the sum of the two numbers indicated by question marks? <table><tr><td>7</td><td>-</td><td>6</td><td>+</td><td>8</td><td>=</td><td>9</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>-</td><td></td><td></td></tr><tr><td>1</td><td>+</td><td>5</td><td>×</td><td>3</td><td>=</td><td>18</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>-</td><td></td><td></td></tr><tr><td>9</td><td>+</td><td>4</td><td>×</td><td>2</td><td>=</td><td>26</td></tr><tr><td>=</td><td></td><td>=</td><td></td><td>=</td><td></td><td></td></tr><tr><td>17</td><td></td><td>15</td><td></td><td>3</td><td></td><td></td></tr></table> <table><tr><td>?</td><td>-</td><td></td><td>+</td><td></td><td>=</td><td>9</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>-</td><td></td><td></td></tr><tr><td></td><td>+</td><td></td><td>×</td><td></td><td>=</td><td>18</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>-</td><td></td><td></td></tr><tr><td></td><td>+</td><td>?</td><td>×</td><td></td><td>=</td><td>26</td></tr><tr><td>=</td><td></td><td>=</td><td></td><td>=</td><td></td><td></td></tr><tr><td>17</td><td></td><td>15</td><td></td><td>3</td><td></td><td></td></tr></table>	7	-	6	+	8	=	9	+		+		-			1	+	5	×	3	=	18	+		+		-			9	+	4	×	2	=	26	=		=		=			17		15		3			?	-		+		=	9	+		+		-				+		×		=	18	+		+		-				+	?	×		=	26	=		=		=			17		15		3		
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39	175 [nickels]	There are 1000 coins in a jar: pennies, nickels, dimes and quarters. The probability of reaching in and randomly selecting a quarter is $\frac{3}{5}$. The probability of reaching in and randomly selecting a dime is $\frac{1}{8}$. Ten percent of the coins are pennies. How many nickels are in the jar? $\frac{3}{5}(1000) = 600$ quarters $\frac{1}{8}(1000) = 125$ dimes $10\% \text{ of } 1000 = 100$ $1000 - 600 - 125 - 100 = 175$ nickels																																																																																																		

40

1014

The positive integers are written in an array according to the following pattern. The first number in each row (in Column 1) indicates how many numbers are in that row. Each entry can be identified by its unique Row and Column number. Find the sum of the Row and Column number for the location of 2026.

Notice that all entries in Column 1 are powers of two, therefore, 2026 will be in Row 11 which has a first number of 1024, and a last number of 2047 which is in column 1024. Count backwards from 2047 to 2026 to discover that 2026 is in Column 1003.

$$1003 + 11 = 1014$$

	Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col. 7	Col. 8	...
Row 1:	1								
Row 2:	2	3							
Row 3:	4	5	6	7					
Row 4:	8	9	10	11	12	13	14	15	
...	...								

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5th Grade

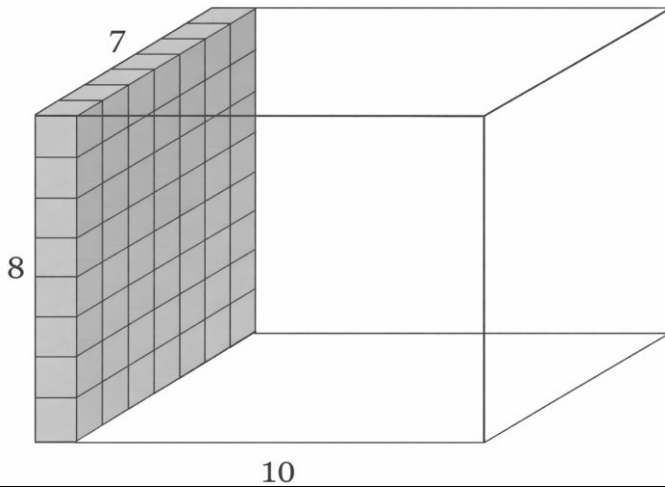
Multiple Choice Solutions

	Answer	Solution			
USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #1 THROUGH #2.					
Mrs. Tamayo has her 5 th grade students keep track of how many minutes of free reading they do each day. The following table shows a sample of four of her students for the week of January 12 th , 2026. All data are given in minutes.					
	Monday	Tuesday	Wednesday	Thursday	Friday
Abinaya	24	30	18	22	?
Jace	32	28	25	30	35
Xizhen	18	20	22	24	36
Neev	40	36	30	28	26

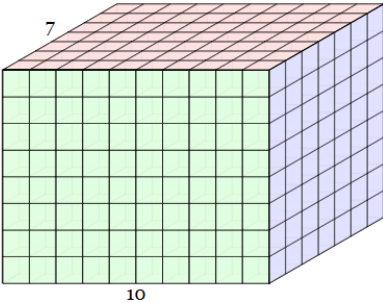
1	C	Abinaya and Xizhen read the same number of minutes for the total week. How many minutes did Abinaya read on Friday? A) 24 B) 25 C) 26 D) 28 E) None of the above. Xizhen: 18 + 20 + 22 + 24 + 36 = 120 Abinaya: 24 + 30 + 18 + 22 = 94. 120 - 94 = 26
2	C	What was the mean number of minutes that the students read for the total week? A) 120 B) 135 C) 137.5 D) 145 E) None of the above. (120 + 150 + 120 + 160)/4 = 550/4 = 137.5

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #3 THROUGH #6.

The following rectangular prism has side lengths of 7, 8 and 10 units, as shown. The prism is being filled with unit cubes.

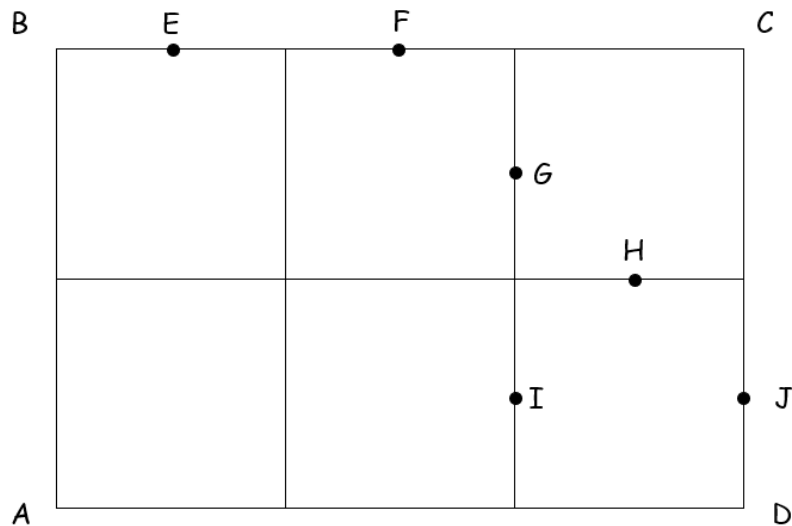


3	A	How many unit cubes are currently in the prism? A) 56 B) 80 C) 87 D) 100 E) None of the above. $8 \times 7 = 56$ unit cubes.
4	C	How many more unit cubes need to be added in order to fill the prism completely? A) 56 B) 240 C) 504 D) 560 E) None of the above. It will hold: $8 \times 7 \times 10 = 560$ cubes $560 - 56 = 504$

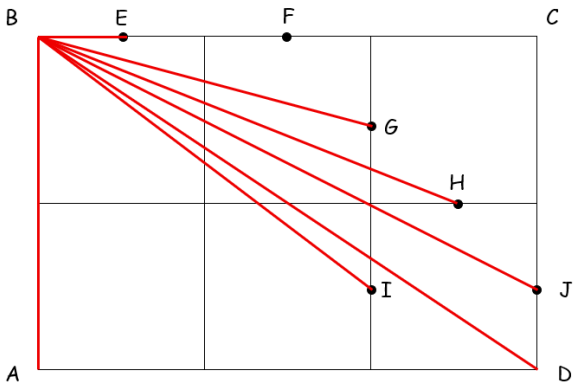
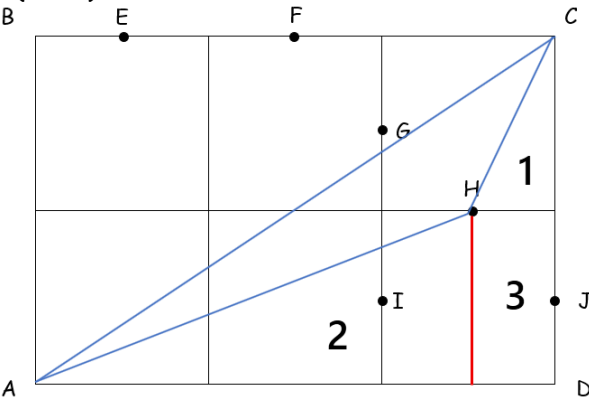
5	B	Once the entire prism is full of unit cubes, all 6 outer surfaces of the prism will be painted. How many unpainted cubes will there be in the interior of the prism? A) 148 B) 240 C) 248 D) 316 E) None of the above. Subtract off the painted cubes: 560 total - 80 front - 80 back This leaves 50 each top and bottom to remove. This leaves 30 left and right. $560 - 160 - 100 - 60 = 240$ interior cubes 
6	D	If all of the cubes with one or more sides painted are put into a bag, and one of them is randomly selected and rolled (like a die), what is the probability that a painted side will be showing on the top after the roll? A) $1/3$ B) $1/2$ C) $421/1280$ D) $103/320$ E) None of the above. There are 320 cubes that have one or more side painted. $320 \times 4 = 1280$ total sides. 8 have 3 painted sides. $5 \times 4 + 8 \times 4 + 6 \times 4 = 76$ with 2 painted sides. $(6 \times 8) \times 2 + (5 \times 6) \times 2 + (5 \times 8) \times 2 = 236$ with 1 painted side. Total painted sides = $(8 \times 3) + (76 \times 2) + 236 = 412$. $P(\text{painted side}) = 412/1280 = 103/320$

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #7 THROUGH #10.

Rectangle ABCD is made up of six congruent (equal) squares. The side length of each square is one unit. The points E, F, G, H, I and J are located at midpoints of the sides of the squares.



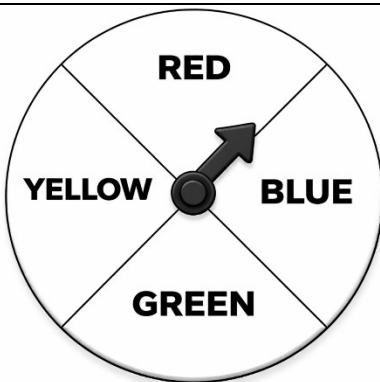
7	C	In units, what is the combined length of all the 1-unit segments that are shown in the figure? A) 15 B) 16 C) 17 D) 18 E) None of the above. There are 3 units in each of the 3 horizontal segments, and there are 2 units in each of the 4 vertical segments. $3 \times 3 + 4 \times 2 = 9 + 8 = 17$
8	E (5)	In units, what is the shortest distance for an ant to crawl from vertex A to vertex C, if the ant must travel along the edges of the squares only. A) 4 B) 6 C) 8 D) 9 E) None of the above. Any path traveling only up and to the right will take 5 units. For example from A to D = 3 units, and from D up to C = 2 units, for a total of 5 units.

9	A	How many different line segments can be drawn from vertex B to one of the other vertices (A, C, D) or to one of the labeled midpoints (E, F, G, H, I, J), as long as the line segment does not pass through another labeled midpoint or vertex? A) 7 B) 8 C) 9 D) 10 E) None of the above. 
10	A	A thrown dart lands randomly anywhere inside rectangle ABCD. What is the probability that it lands in triangle ACH? A) 1/6 B) 2/25 C) 10% D) 12% E) None of the above. For simplicity, assume that each square has a side length of 4. The area below AC is $(12)(8)/2 = 48$. Subtract off the area of regions 1 (triangle), 2 (triangle below AH), and 3 (rectangle). 1: $\frac{1}{2}(2)(4) = 4$ 2: $\frac{1}{2}(10)(4) = 20$ 3: $(2)(4) = 8$ Area ACH = $48 - 4 - 20 - 8 = 16$ $P(ACH) = 16/96 = 1/6$ 

“Math is Cool” Championships -- 2025-26

5th Grade

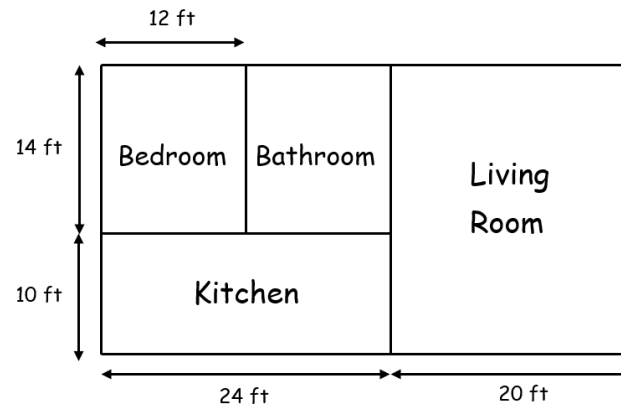
Team Test Solutions

	Answer	Solution
1	2,940 [Labubus]	A Labubu toy factory makes 84 Labubus in 1 hour. If they work 7 hours a day, how many Labubus do they make in 5 days? $84 \times 7 \times 5$
2	41 [= sum]	What is the sum of the prime numbers between 1 and 15? 2, 3, 5, 7, 11, 13 $2+3+5+7+11+13 = 41$
3	256 [square feet]	A square garden has a perimeter of 64 feet. In square feet, what is the area of the garden? $64 / 4 = 16$ feet per side of the square $16 \times 16 = \text{Area} = 256$ square feet
4	25 [times]	A circular spinner is divided into 4 equal sections: red, blue, green, and yellow. When it is spun 100 times, what is the expected number of times the spinner lands on green? The spinner should be expected to land on green $\frac{1}{4}$ times, since the spinner is broken up into 4 equal parts. $\frac{1}{4}$ of 100 = 25. 
5	21	Evaluate: $5(4 + 2) - 3(1 + 2)$ $5(4 + 2) - 3(1 + 2) =$ $5(6) - 3(3) =$ $30 - 9 = 21$
6	75 [gallons]	A tank can hold 180 gallons of water. It is $\frac{2}{3}$ full. After some water is drained, it is $\frac{1}{4}$ full. How many gallons were drained? $\frac{2}{3}$ of 180 is 120. It starts at $\frac{2}{3}$ full, meaning there are 120 gallons to begin with. Then, it drains until it gets $\frac{1}{4}$ left. $\frac{1}{4}$ of 180 is 45. $120 - 45 = 75$. The tank drained 75 gallons.

7	20 [= sum]	The following pyramid is an example of a "self-descriptive" sequence, where each row describes the digits that are in the previous row. For example, the 2nd row tells us that there was one '2' in the first row. The 3rd row tells us that there was one '1' and one '2' in the second row. A total of 6 rows of the pyramid are included. What is the sum of the digits in the 7th row? <pre> 2 1 2 1 1 1 2 3 1 1 2 1 3 2 1 1 2 1 1 1 3 1 2 2 1 1 2 </pre> The 7th row has three 1s, one 3, one 1, two 2s, two 1s and one 2. $3 + 1 + 1 + 3 + 1 + 1 + 2 + 2 + 2 + 1 + 1 + 2 = 20$ <pre> 2 1 2 1 1 1 2 3 1 1 2 1 3 2 1 1 2 1 1 1 3 1 2 2 1 1 2 3 1 1 3 1 1 2 2 2 1 1 2 </pre>
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8**35 [%]**

Given the following floor plan of Abigail's new apartment, the area of the bathroom is what percentage of the area of the living room?



The living room is $20 \times 24 = 480$ sq feet. The bathroom is $12 \times 14 = 168$ sq feet.

The ratio is $168/480$ which simplifies to $7/20 = 35/100$, or 35%.

9

50 [ways]
[squares]

Start with the straight up-and-down squares, which are straightforward:

1x1: 16

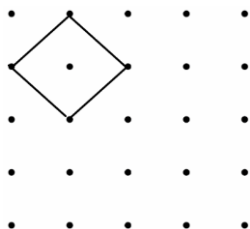
2x2: 9

3x3: 4

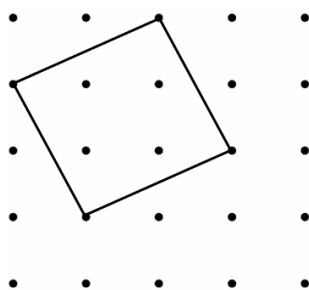
4x4: 1

Now move on to the tilted squares:

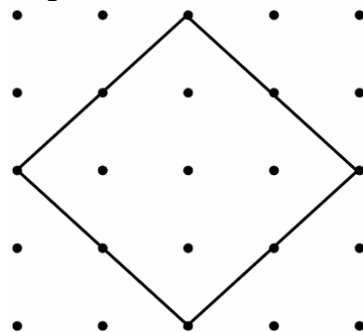
Small: 9



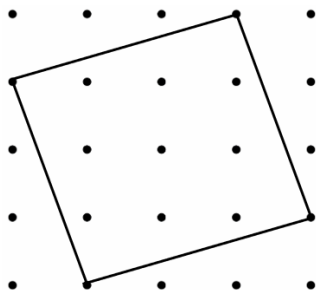
Medium: 8 total, 4 tilted in the direction shown here, and 4 tilted in the other direction.



Large: 1



Largest: 2 (as shown, and tilted in the other direction).



Total = 50

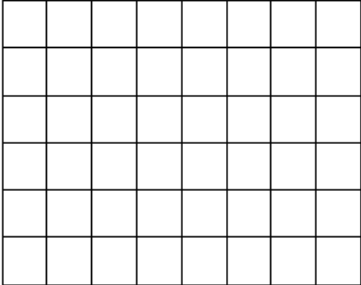
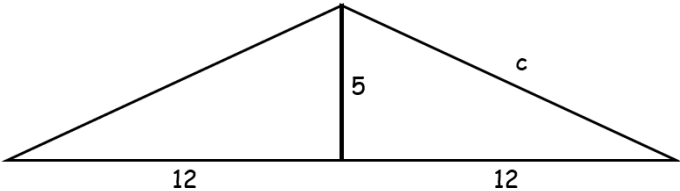
10	900 [bricks]	Two bricklayers are building a wall. Sarah would take 9 hours to build the wall alone, and Murphy would take 10 hours to build it alone. Working together, they talk too much, which causes their combined output to be reduced by 10 bricks per hour. Together, they build the wall in 5 hours. How many bricks are in the wall? Assume x bricks. S: 9 hours/x bricks M: 10 hours/x bricks Together, assuming no slowdown: T: $(9 \times 10)/(9+10) = 90/19$ hours per x bricks = $4 \frac{14}{19}$ hours per x bricks It actually took 5 hours, and at 10 less bricks per hour that is 50 bricks. Set up a ratio: $90/19$ hours / x bricks = 5 hours / $(x+50)$ bricks $90/19 (x + 50) = 5x$ $90x + 4500 = 95x$ $5x = 4500$ $X = 900$ bricks
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"Math is Cool" Championships -- 2025-26

5th Grade

Linda Moore Triple Jump Solutions

	Answer	Solution
1	60 [dollars]	Natalie has 2 dozen cookies. She sells each cookie for \$2.50. In dollars, how much money will she make if she sells every cookie? 1 dozen = 12 2 dozen = 24 $24 \times \$2.5 = \60
2	7 [centimeters]	In centimeters, what is the diameter of a circle that has a radius of 3.5 centimeters? Diameter = $2 \times \text{radius}$ $3.5 \times 2 = 7$
3	75 [percent]	A bag has 5 red marbles, 4 blue marbles, and 3 green marbles. One marble is chosen at random. What is the probability, expressed as a percentage, that it is not green? We want the probability of choosing a marble that is any color but green. There are a total of 12 marbles, and 9 are not green. $9/12 = 3/4 = 75\%$
4	663 [= the no.]	A three-digit positive integer has digits that add to 15. The hundreds digit is twice the ones digit. The tens digit equals the hundreds digit. What is the number? Can list the possibilities: 221, 442, 663, 884 Only 663 has digits that add to 15.
5	144 [blocks]	A staircase is built with blocks so that step 1 has 1 block, step 2 has 3 blocks, step 3 has 5 blocks, and so on. If the staircase has 12 steps, how many blocks are needed in total? Step n uses the first n odd numbers pattern: step 1 =1, step 2 =3, step 3 =5, ... For step 12, the total is $n \times n$. $12 \times 12 = 144$. Or, just add the numbers using brute force: $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 = 144$

6	36 [markers]	Mr. Savage has 4 markers for every 9 pencils in his classroom. If there are 45 more pencils than markers, how many markers are there? 4 markers : 9 pencils Ratio up by a factor of 9 to get: 36 markers : 81 pencils, where $81 - 36 = 45$.
7	47 [breaks]	A chocolate candy bar has dimensions of 6 by 8 units. What is the minimum number of breaks, either horizontal or vertical, that are needed to separate it into pieces of 1 by 1 units? Pieces cannot be stacked on top of each other before breaking. Start by making 5 horizontal breaks, to separate into 6 strips of 8 pieces each. Each strip now needs 7 breaks. Total breaks = $5 + 6 \times 7 = 47$. 
8	13 [inches]	The base of an isosceles triangle is 24 inches, and its area is 60 square inches. What is the length in inches of one of the congruent sides? Area = $\frac{1}{2}bh$ $60 = \frac{1}{2}(24)h$ Solve for $h = 5$ Use the Pythagorean Theorem: $12^2 + 5^2 = c^2$ $144 + 25 = c^2$ $169 = c^2$ $C = 13$ 

9	32 [subsets]	How many subsets of the set of numbers $\{1, 2, 3, 4, 5, 6, 7, 8\}$ contain both 1 and 2, but do not contain 7? Assume that 1 and 2 are included. Throw out the 7, so there are 5 remaining items to choose from: 3, 4, 5, 6, 8. Only 1 and 2: 1 subset 1 and 2 + one other: 5 1 and 2 + 2 others: 10 (can list them, or do $5C2$). 1 and 2 + 3 others: 10 (equivalent of doing 2 others) 1 and 2 + 4 others: 5 (exclude one each time) 1 and 2 + 5 others: 1 Total = $1 + 5 + 10 + 10 + 5 + 1 = 32$
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10	27 [= sum]	From a standard deck of playing cards, all 13 hearts have been lined up in a row, with positions 1 through 13, as indicated below. Face cards are defined as Jack, Queen and King. The Ace is the high card, so the natural order of the cards from left to right is 2 through 10, Jack, Queen, King and Ace. Assume a value of 15 for the ace and face cards. Otherwise, assume their numerical value. Given the following clues, determine the order of the 13 cards. What is the sum of the card values in positions 5, 7 and 8? Clues: • No two cards are in their natural order from left to right. • No face cards are at either end. • Two face cards are adjacent. • The King and Ace are separated by one card. • The 2nd and 3rd cards have a sum of 13. • The 10th and 11th cards have a sum of 10. • The 8 is two places to the left of the ace. • The Jack is two places to the right of the 2. • The 4 is two places to the right of the 3. • The 6 is three places to the left of the ace. • The 9 is somewhere to the right of the 4. • The 10 is two places to the right of the 5. • The third card from the right and the third card from the left are both odd. See solution below. 5th: 10 7th: 2 8th: Q 10 + 2 + 15 = 27 <table><tr><td><u>6</u></td><td><u>8</u></td><td><u>5</u></td><td><u>A</u></td><td><u>10</u></td><td><u>K</u></td><td><u>2</u></td><td><u>Q</u></td><td><u>J</u></td><td><u>3</u></td><td><u>7</u></td><td><u>4</u></td><td><u>9</u></td></tr><tr><td>1st</td><td>2nd</td><td>3rd</td><td>4th</td><td>5th</td><td>6th</td><td>7th</td><td>8th</td><td>9th</td><td>10th</td><td>11th</td><td>12th</td><td>13th</td></tr></table>	<u>6</u>	<u>8</u>	<u>5</u>	<u>A</u>	<u>10</u>	<u>K</u>	<u>2</u>	<u>Q</u>	<u>J</u>	<u>3</u>	<u>7</u>	<u>4</u>	<u>9</u>	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th	9 th	10 th	11 th	12 th	13 th
<u>6</u>	<u>8</u>	<u>5</u>	<u>A</u>	<u>10</u>	<u>K</u>	<u>2</u>	<u>Q</u>	<u>J</u>	<u>3</u>	<u>7</u>	<u>4</u>	<u>9</u>																
1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th	9 th	10 th	11 th	12 th	13 th																

"Math is Cool" Championships -- 2025-26

5th Grade

College Bowl Round #1 Solutions

#	Problem	Answer
1	One hundred twenty-three plus what number equals two hundred? $200 - 123 = 77$	77
2	A train leaves at nine forty-five AM and arrives at its destination at one twenty PM on the same day. How many minutes did the trip take? $9:45$ to $12:45 = 3\text{hrs} = 180$ minutes From $12:45$ to $1:20$ there are 35 minutes $180 + 35 = 215$ minutes total	215 [minutes]
3	What is eight hundred fifty-two point four nine rounded to the nearest ten? 852.49 rounds down to 850.	850
4	How many different ways are there to arrange the digits in the number two thousand twenty-six that do not have a zero in the first position? There are $4!/2! = 12$ total ways, but 3 of them have a leading zero (0622, 0262, 0226). $12 - 3 = 9$	9 [ways]
5	What is the minimum number of digits that must be removed from the number twelve million three hundred twenty-three thousand three hundred fourteen so that the remaining digits are a palindrome? 12323314 12321 Remove the 3, 3 and 4, and the result is a palindrome.	3 [digits]
6	Leo is going to fill a box with measurements forty by forty by sixty units, using identical cubes of some size. What is the minimum number of cubes needed? If the cubes are 20x20 units, a 2x3 array will fill the box.	12 [cubes]
7	A movie theater has twelve rows of seats with eighteen seats in each row. One show sold one hundred sixty-eight tickets. How many empty seats were there? $12 \times 18 = 216$ $216 - 168 = 48$ seats	48 [seats]

8	What is the next number in the pattern: Five, ten, twenty, forty, and so on Multiplying by two each time.	80
9	Mia's water bottle can hold twenty-four ounces of water. She drinks two full water bottles each day. How many total cups of water does Mia drink each day? 8 ounces = 1 cup 24 ounces = 3 cups She drinks two 24oz of water a day. $3 \times 2 = 6$	6 [cups]
10	A book has two hundred fifty-six pages. If a student reads thirty-two pages each day, how many full days will it take to finish the book? $256/32 = 8$	8 [days]

“Math is Cool” Championships -- 2025-26

5th grade

College Bowl Round #2 Solutions

#	Problem	Answer
1	What is the smallest 3-digit integer sum of two prime numbers? $97 + 3 = 100$	100
2	What is the next number in the following sequence? Seventeen, eleven, five, negative one, and so on. Subtract 6 each time. $-1 - 6 = -7$.	-7

3	Milo, Nim, Tyler and Lacy took turns in that order saying the whole numbers from one to twenty-five. How many times did Milo say a number that contains the digit one? Milo said: 1, 13, 17, and 21. <table><tr><td>1</td><td>M</td></tr><tr><td>2</td><td>N</td></tr><tr><td>3</td><td>T</td></tr><tr><td>4</td><td>L</td></tr><tr><td>5</td><td>M</td></tr><tr><td>6</td><td>N</td></tr><tr><td>7</td><td>T</td></tr><tr><td>8</td><td>L</td></tr><tr><td>9</td><td>M</td></tr><tr><td>10</td><td>N</td></tr><tr><td>11</td><td>T</td></tr><tr><td>12</td><td>L</td></tr><tr><td>13</td><td>M</td></tr><tr><td>14</td><td>N</td></tr><tr><td>15</td><td>T</td></tr><tr><td>16</td><td>L</td></tr><tr><td>17</td><td>M</td></tr><tr><td>18</td><td>N</td></tr><tr><td>19</td><td>T</td></tr><tr><td>20</td><td>L</td></tr><tr><td>21</td><td>M</td></tr><tr><td>22</td><td>N</td></tr><tr><td>23</td><td>T</td></tr><tr><td>24</td><td>L</td></tr><tr><td>25</td><td>M</td></tr></table>	1	M	2	N	3	T	4	L	5	M	6	N	7	T	8	L	9	M	10	N	11	T	12	L	13	M	14	N	15	T	16	L	17	M	18	N	19	T	20	L	21	M	22	N	23	T	24	L	25	M	4 [times]
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4	The length of a rectangle is eight inches, and the width is half as long. In inches, what is the side length of a square with the same perimeter as the rectangle? The rectangle dimensions are 8x4, so the perimeter is $12 \times 2 = 24$. Therefore the side length of the square is $24 / 4 = 6$	6 [in]																																																		
5	X plus X plus six equals X plus X plus X plus X. What is the value of X plus two? $2x + 6 = 4x$ $2x = 6$ $x = 3$ $x + 2 = 5$	5 [= x + 2]																																																		

6	The sum of the digits of a six-digit whole number is five. What is the product of the digits? There must be at least one 0 in the number, for example 111,110 meets the criteria. The product will always be 0.	0 [= product]
7	Julian had fifty dollars saved. He bought two board games that each cost eighteen dollars. Then he earned twelve dollars by doing chores. How many dollars does Julian have now? $18 (2) = 36$ $50 - 36 = 14$ $14 + 12 = 26$ dollars	26 [dollars]
8	In a line of students, there are twenty-seven students in front of Ivy and thirteen students behind her. Stefan is exactly in the middle. How many students are between Ivy and Stefan? Ivy is in position #28 out of 41 total students, and Stefan is in position #21. There are 6 students between them.	6 [students]
9	A rectangle is sixteen inches long and three inches wide. In square inches, what is one-half of its area? Area is length x width $16 \times 3 = 48$ $48/2 = 24$	24 [square inches]
10	A pizza was cut into twelve equal slices. Three friends each ate three slices. What percent of the pizza is left? $3 \times 3 = 9$ slices eaten, 3 slices left. $3/12 = \frac{1}{4} = 25\%$	25 [%]

“Math is Cool” Championships -- 2025-26

5th grade

College Bowl Round #3 Solutions

#	Problem	Answer
1	Rachel buys three notebooks for four dollars each and two pens for two dollars each. She pays with a twenty dollar bill. How many dollars does she get back? $3 \times 4 = 12$ $2 \times 2 = 4$ $12 + 4 = 16$ $20 - 16 = 4$ dollars remaining	4 [dollars]
2	The area of a rectangle is forty-eight square units. If the width is six units, what is the length in units, times two? Area = length $\times$ width $48 = (x)(6)$ $48 \div 6 = x$ $x = 8, 2(8) = 16$	16 [units]
3	A ten-sided die is numbered zero through nine. On a single roll, what is the probability in percent of rolling a number less than four? Outcomes: 0, 1, 2, 3 are less than 4. $4/10 = 40\%$	40 [%]
4	The first three terms in an arithmetic sequence are: twenty-one halves, eleven, and twenty-three halves. What is the 6th term? $21/2, 11, 23/2, 12, 25/2, 13$ Add $\frac{1}{2}$ each time.	13 [= 6 th term]
5	Using the digits one through eight exactly once each, form two four-digits numbers with a sum as small as possible. What is the sum? Use the smallest digits for the larger place values: $1458 + 2367 = 3825$.	3825 [= sum]
6	A standard deck of playing cards has the Jacks, Queens and Kings removed from the deck. If one card is randomly selected, what is the probability in percent that it is an ace or a two? 52 cards total - 12 face cards = 40 cards remaining. There are 4 aces and 4 twos. $8/40 = 0.2 = 20\%$	20 [%]

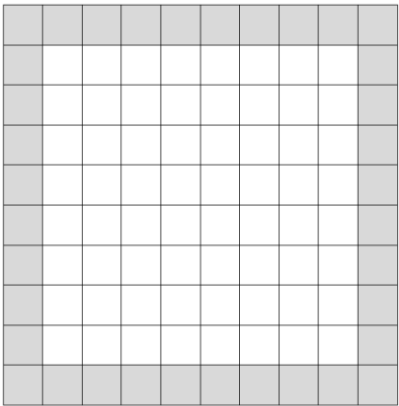
7	A group of students spend the day reading. Three students read for twenty minutes, four for thirty minutes, and three for forty minutes. How many hours did the students spend reading in total? $3(20) = 60$ minutes $4(30) = 120$ minutes $3(40) = 120$ minutes $60 + 120 + 120 = 300$ minutes $300 \div 60 \text{ minutes/hour} = 5 \text{ hours}$	5 [hours]
8	Solve for X: Four X plus twelve equals forty-four $4x + 12 = 44$ $4x = 32$ $x = 8$	8 [=x]
9	How many total sides do three octagons, two hexagons and one trapezoid have? $3 \times 8 + 2 \times 6 + 4 = 24 + 12 + 4 = 40$	40 [sides]
10	A pizza costs fifteen dollars. How many can you buy with ninety dollars? $90 \div 15 = 6$	6 [pizzas]

"Math is Cool" Championships -- 2025-26

5th grade

College Bowl Round #4 Solutions

#	Problem	Answer
1	A recipe for chocolate chip cookies calls for two-thirds cup of sugar. How many cups of sugar is needed if Jenny wants to triple the recipe? $\frac{2}{3} \times 3 = 2$	2 [cups]
2	Solve for X: Twenty minus X equals seven minus eight $20 - x = 7 - 8$ $20 - x = -1$ $X = 21$	21 [= x]
3	Connor scored eight hundred two thousand one hundred fourteen points playing an arcade game. Esmat scored seven hundred ninety-eight thousand one hundred ten points in the game. How many more points did Connor score? $802,114 - 798,110 = 4,004$	4,004 [points]
4	The mean of the following numbers is twenty-four. What is the value of X? X, twelve, fourteen, forty If mean = 24, sum = $24 \times 4 = 96$. $12 + 14 + 40 = 66$ $96 - 66 = 30$.	30 [= x]

5	Jose built a square outer wall using thirty-six identical unit cubes. How many unit cubes are needed to completely fill the interior of the square? The outer wall will have dimensions 10x10. The next layer in is 8x8, so 64 cubes are needed. 	64 [cubes]
6	How many whole numbers X are there for which X minus twenty-four and X plus twenty-four are two-digit numbers? At a minimum, X can equal 34, and at a maximum X can equal 75. $75 - 33 = 42$ numbers.	42 [whole numbers]
7	A sequence begins with: One, negative one, negative one, one, negative one, and so on. Each number from the third on is equal to the product of the two preceding numbers. What is the sum of the first ten numbers? 1, -1, -1, 1, -1, -1, 1, -1, -1, 1 Sum = $4 - 6 = -2$	-2 [= sum]
8	Rafa rolled a six-sided die four times, and scored a total of twenty-three points. How many sixes did he roll? The only way to get 23 is with: 6, 6, 6, 5.	3 [sixes]
9	Sophie and her mom want to rent a canoe. Canoe rentals cost forty-five dollars for a half hour. If Sophie and her mom rent a canoe for one hour and thirty minutes, how many dollars will it cost? $1/2$ hour (30 minutes) = \$45 1 hour = \$90 1 hour + 30 minutes = $\\$90 + \\$45 = \\$135$	135 [dollars]
10	What is two-thirds of two hundred seventy? $2/3 * 270 = 180$	180

"Math is Cool" Championships -- 2025-26

5th grade

College Bowl Round #5 Solutions

#	Problem	Answer
1	What is one hundred forty-four times three-fourths? $144 \times \frac{3}{4} = 108$	108
2	The side lengths of a triangle are six, ten and eleven units. An equilateral triangle has the same perimeter as this triangle. How long is one side of the equilateral triangle, in units? $6 + 10 + 11 = 27$ $27/3 = 9$	9 [units]
3	The third through fifth terms of an arithmetic sequence are one-half, three-fourths, and one. What is the first term? $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$. The common difference is $\frac{1}{4}$	0 [= 1 st term]
4	How many of the whole numbers from one through nine is the following number divisible by? Two million three hundred ninety-two thousand six hundred forty-three 2,392,642 is only divisible by 1.	1 [number]
5	Ian has a stapler, a ruler, a calculator and a compass. How many different ways can he line them up on his desk if the calculator and compass both have to be next to the stapler? Four objects: A, B, C, D B and C both have to be next to A, so the only options are: DBAC BACD DCAB CABD	4 [ways]
6	What is the units digit when the first five odd whole numbers are multiplied together? $1 \times 3 \times 5 \times 7 \times 9$ Because it includes a 5, but not a 2, it's going to end in 5.	5 [= units digit]

7	Maria has forty-nine blue pencils and one red pencil. How many blue pencils does she have to get rid of so that ninety percent of her pencils are blue? There's only one red, and if she had 9 blue that would be 90%. Therefore, get rid of 40 to have 9 remaining.	40 [pencils]
8	Packard is twenty-one years old. In years, what is the mean of his age seven years ago and his age eleven years ago? $21 - 7 = 14$ $21 - 11 = 10$ $(14 + 10)/2 = 12$	12 [years]
9	Two million equals two hundred times what number? $2000000/200 = 10000$	10,000 [ten thousand]
10	Ten runners are in a cross-country race. At the finish, there are three more runners behind Oliver than there are in front of him. In what position number did Oliver finish? In 4th, there are 3 people ahead of him and 6 people behind him	4 [or 4 th]

"Math is Cool" Championships -- 2025-26

5th grade

College Bowl Round #6 Solutions

#	Problem	Answer
1	A crate is 6 feet long, 24 inches wide, and 3 feet tall. What is the volume in cubic feet? Volume = length x width x height Volume = $6 \times 2 \times 3 = 36$ feet cubed	36 [feet cubed]
2	Find the value of x that makes the expression true. X plus fifteen equals thirty plus two $X + 15 = 30 + 2$ $X = 32 - 15 = 17$	17
3	What is the product of seven, three and five? Product is the answer to a multiplication problem $7 \times 3 \times 5 = 105$	105 [= product]
4	June has some red and some black marbles in a bag, with a total of twenty-five marbles. If one marble is randomly selected, the probability that it is red equals X and the probability that it is black equals four X. How many black marbles are there? The probabilities must add to 100%, so $x + 4x = 5x = 100$, $x = 20\%$ chance of red, and 80% chance of black. $20/25 = 80/100$, so there are 20 black marbles.	20 [black marbles]
5	How many times more even divisors than odd divisors does the number thirty-six have? 36 has 9 divisors. Six are even and 3 are odd (1, 3, 9), therefore $6/3 = 2$ times more.	2 [times]
6	A bicyclist travels five meters in one second. The wheels on the bike have a circumference of one hundred twenty-five centimeters. How many complete revolutions does each wheel make in five seconds? $5 \text{ m} = 500 \text{ cm} / 125 \text{ cm} = 4$ revolutions per second. $4 * 5 = 20$	20 [revolutions]

7	Alex has a one-meter-long rope and a two-meter-long rope. He cuts up both ropes so that the resulting pieces are all of equal length. What is the greatest number of pieces less than twenty that he can obtain in this manner? The numbers will all be multiples of 3, because he is starting with 3 one-meter sections. The largest multiple of 3 less than 20 is 18.	18 [pieces]
8	A sequence begins with zero, five, ten, fifteen, and so on. What is the sum of the first eight numbers in the sequence? $0 + 5 + 10 + 15 + 20 + 25 + 30 + 35 = 140$	140 [= sum]
9	How many inches are in nine point two five feet? 12 inches in 1 foot $12 \times 9 = 108$ The additional $\frac{1}{4}$ foot adds 3 inches.	111 [inches]
10	What is the perimeter of a square divided by the sum of two side lengths of the square? The perimeter of a square = $4s$. $4s/2s = 2$.	2